

Computer Network | 2301CS501 | Lab Planning

No	PlanningTitle	PlanningDescription
1	Study of basic networking commands and IP configuration.	1. ping 2. ipconfig / ifconfig 3. getmac 4. hostname 5. systeminfo 6. nslookup 7. tracert / traceroute 8. pathping 9. netstat 10. arp 11. dig
2	Study of various networking devices.	1. Hub 2. Switch 3. Router 4. Bridge 5. Gateway 6. Modem (DSL and ADSL) 7. Ethernet Card
3	Study of Transmission Media, Connectors and Crimping a LAN.	A. Study different types of network cables & connectors and practically implement the cross-wired cable using a crimping tool. B. Study different types of network cables & connectors and practically implement the straight-through cable using a crimping tool.
4	Installation of Network Simulator (Packet Tracer) and Implement different LAN topologies.	1. Point-to-Point Connection and Client-Server Connection 2. Bus 3. Ring 4. Star 5. Tree 6. Mesh
5	Study the concept of VLAN using packet tracer.	Implement the different network structures in VLAN and multiple VLAN trunking.
6	To configure Telnet & SSH for remote connection.	To configure Telnet and SSH services on a network topology, perform remote login operations, and capture network packets to analyze security differences (plain text vs. encryption).
7	Study the application layer protocol DNS, DHCP, FTP.	Create a simple network and implement the application layer protocol DNS, DHCP & FTP using packet tracer
8	Study & survey of Darshan University network infrastructure.	Visit our university server room to know how to manage switch and vlan concept with various networks and collect information about working of port forwarding and firewall.
9	Study Packet capture and header analysis by Wireshark.	Study Packet capture and header analysis by Wireshark. A. Capturing and observe Live Traffic – Use any website. B. Applying Capture Filters: a. HTTP traffic: http i. For HTTP - Inspect request and response headers, URLs, and status codes. ii. For HTTPS, observe the handshake process (TLS) . b. DNS requests: dns i. Observe how domain names are translated into IP addresses. ii. Identify DNS request and response packets. c. TCP packets: tcp i. Observe the 3-way handshake (SYN, SYN-ACK, ACK). ii. Identify sequence numbers and acknowledgment numbers. d. ICMP (ping): icmp i. Send ping requests (ping google.com) and observe request/reply packets.
10	Study of IP Addressing and sub-netting.	A. Find default subnet masks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address. a) 8.1.4.5/16 b) 130.4.102.1/24 c) 199.1.1.1/24 d) 130.4.102.1/22 e) 199.1.1.100/27 B. Determining the Subnet for Devices A and B: - Device A: 172.16.17.30/20 - Device B: 172.16.28.15/20 [Extra sum given in pdf]

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11	Study the concept of dynamic routing using packet tracer.	A. Connect the two different networks based on the given IP addresses and subnet using a packet tracer. B. Connect the three different networks based on the given IP addresses and subnet using a packet tracer.
12	Study the concept of static routing using packet tracer.	A. Connect the two different networks based on the given IP addresses and subnet using a packet tracer. B. Connect the three different networks based on the given IP addresses and subnet using a packet tracer.
13	Study and Configure Network Address Translation.	To configure Static NAT and Port Address Translation (PAT) on a Cisco Router and verify communication between private and public networks.
14	To study load balancer in different network scenario.	A. To configure loadbalancing with static route using packet tracer. B. To configure reverse proxy using nginx. C. To configure load balancer using nginx.
15	Understand how containers communicate using different network drivers and learn to troubleshoot connectivity issues using Docker networking commands.	Create a Docker-based three-tier application environment consisting of: 1) Frontend Container (Nginx) 2) Backend Container (Node.js) 3) Database Container (MySQL) - Create and inspect a Bridge network. - Launch multiple containers and verify communication between them. - Configure Host networking and compare its behavior with Bridge networking. - Create an Overlay network (Docker Swarm mode) and deploy containers on it. - Configure port mapping and access containerized services from the host. - Use networking troubleshooting commands and Identify and resolve connectivity issues such as: Incorrect network attachment, Missing port mapping, DNS resolution problems between containers, Service accessibility issues